ZK-PoC device probe

Measures achievable F(d) — GFLOPS through the browser — on the WebGPU and CPU paths, at a controlled duty cycle. Output feeds DEFAULT_TIERS in bench/breakeven.py, replacing the literature-anchored placeholders.

1 · Device

user agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
logical cores	8
device memory	16 GB (coarse)
WebGPU	available
GPU adapter	intel / gen-12lp
battery	33% (discharging)

2 · Benchmark

matrix N 512 duty cycle 100% (peak F(d)) reps 7

Run probe Copy JSON done

3 · Results

path	achievable F(d)	median	check
CPU (JS typed array)	0.58 GFLOPS	462.2 ms	—
GPU (WebGPU / WGSL)	38.35 GFLOPS	7.0 ms	verified (rel err 0.0e+0)
config	N=512, duty=100%, $2N^3$=0.27 GFLOP/rep		
battery drop	0.00 % over the run		

4 · Export


```
{
  "schema": "zkpoc.device-probe/1",
  "captured_at": "2026-08-03T05:56:50.718Z",
  "config": {
    "matrix_n": 512,
    "duty_cycle": 1,
    "reps": 7
```

```
    },
    "device": {
      "ua": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1
      "platform": "Win32",
      "cores": 8,
      "memory_gb": 16,
      "webgpu": true,
      "adapter": {
        "vendor": "intel",
        "architecture": "gen-12lp",
        "device": null,
        "description": null
      },
      "battery": {
        "level": 0.33,
        "charging": false,
        "dischargingTime": 2106
      },
      "limits": {
        "maxComputeWorkgroupStorageSize": 32768,
        "maxBufferSize": 2147483648
      }
    },
    "battery_after": {
      "level": 0.33,
      "charging": false,
      "dischargingTime": 2106
    },
    "results": {
      "cpu": {
        "path": "js-typed-array",
        "median_ms": 462.20000000298023,
        "min_ms": 408.8999999985099,
        "max_ms": 706.7000000029802,
        "samples": [
          408.8999999985099,
          425.1000000014901,
          444.90000000596046,
          462.20000000298023,
          533.8999999985099,
          703.7000000029802,
          706.7000000029802
        ],
        "gflops": 0.5807777066167658
      },
      "gpu": {
        "path": "webgpu-wgsl",
        "median_ms": 7,
        "min_ms": 5.799999997019768,
        "max_ms": 10.800000004470348,
        "samples": [
          5.799999997019768,
          6.299999997019768,
          6.399999998509884,
          7,
          7.800000004470348,
```

```
        8.799999997019768,
        10.800000004470348
      ],
      "gflops": 38.34792228571429,
      "correct": true,
      "rel_error": 0
    }
  },
  "note": "cpu row is a JS lower bound, not WASM SIMD; see M1"
```


Scope. The CPU row uses a JS typed-array kernel, which is a defensible lower bound but not the Rust→WASM SIMD path — that kernel lands in M1 and will replace this row. The WebGPU row is the one that matters: the break-even model shows the CPU path is economically dead regardless.

Energy. Browsers expose no power draw. Battery Status is sampled where available, but real W(d) needs host-side RAPL / Intel Power Gadget / powermetrics — see bench/power/. Quantifying energy and thermal effects is the gap FibRace (arXiv:2510.14693 §5.1.3) explicitly left open, so it is instrumented deliberately rather than assumed.